

DONALD LAISHRAM

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Skills

Python | **SQL** | **C** | **C#** | **Java** | **.NET** | PyTorch | TensorFlow | Scikit-Learn | Hugging Face | Scipy | Apache Spark | Model Training | Conformer | LLM Integration | LangChain/LangGraph | MCP | Anthropic API | RAG | Pinecone | Weaviate | Ollama | MLflow | ETL | Data Pipelines | Databricks | Docker | CI/CD | AWS | Azure | FastAPI | Flask | REST APIs | Microservices | PostgreSQL | MySQL | GraphDB | Git | GitHub | Unit Testing | Robot Framework | Web Scraping | Streamlit | Jupyter | VS Code | Cursor

Experience

Associate AI Engineer | Datamatics | Mumbai | Jan 2025 - Present

Engineered and productionised distributed data pipelines utilizing **Databricks**, **Apache** frameworks, and **SQL (SSMS)** to **orchestrate** the end-to-end extraction, deterministic transformation, algorithmic validation, and feature preparation of multi-modal enterprise datasets for downstream Machine Learning architectures.

Developed **AI workflows** involving **model fine-tuning and optimization**, incorporating parameter-efficient fine-tuning (**PEFT/LoRA**) and post-training quantization via **Hugging Face** ecosystems; pioneered automated **MLflow** pipelines to govern cross-validation experimentation, programmatic model evaluation, artifact versioning, and high-throughput production serving.

Developed **full-stack AI APIs** using Python/FastAPI and REST APIs, integrating **OpenAI/Ollama** LLM services with databases and existing enterprise applications for production.

Implemented **MLOps and model deployment pipelines** using **MLflow and Docker**, supporting model lifecycle management, containerized deployment, and maintenance of production data/ML workflows.

Software Engineer Intern | NTRO | New Delhi, IN | Sept 2024 - Nov 2024

Built an NLP summarization pipeline (BART, Longformer) over a self-collected 2M+ word dataset, achieving ROUGE-1 F1 scores up to 0.5 on real-time report summarization up to 1.2M words, deployed via Docker.

Developed GraphDB/SPARQL APIs (Python Flask) to query RDF data in JSON-LD format, and built a search-engine-style interface handling up to 1,000 queries/second.

Software Engineer Intern | Cloudstrats | Mumbai, IN | Jun 2023 - Dec 2023

Developed ASR and Speech AI technologies, specializing in noise cancellation and model training for robust speech recognition. Integrated the latest Conformer-2 architecture to improve ASR performance and led data augmentation for enhanced accuracy.

Projects

O2C- Collections (Automated AI calling)

Engineered a FastAPI + OpenAI to process call recording JSON payloads from the IVR/dialer system (Marco), generating automated summaries of customer conversations and auto-updating payment status and Promise-to-Pay (PTP) dates in the Finato platform, reducing manual follow-up review effort.

Conformer — E2E Speech Recognition custom model

Implemented a custom **Conformer architecture** combining multi-head self-attention and depthwise convolution modules for end-to-end ASR. Trained the model on **~960 hours** of the LibriSpeech corpus (OpenSLR) using TensorFlow/Keras and Google Colab TPU, with custom STFT-based audio preprocessing, character-level tokenization, autoregressive decoding, and warm-up/decay learning-rate scheduling. Achieved **15% WER** on the LibriSpeech test-clean benchmark

RAG-Based RDF Knowledge Retrieval using GraphDB

Engineered APIs using GraphDB, Python Flask, SPARQL, and JSON-LD to query and retrieve structured military-domain knowledge. Developed a **RAG-based retrieval layer** and search-engine-style interface for semantic exploration and visualization of RDF entities and relationships, enabling contextual retrieval for downstream AI applications.

Achievements

Smart India Hackathon 2023 Finalist | Multilingual ASR Project 19&20 Dec 2023

Developed a multilingual ASR model to support five Indian languages, achieving finalist status. Recognized for innovation and commitment to technological advancement